

INPUT

10 stereo pairs
ChArUco board

Rayfield Measurement (2D)

Ray2D Preprocessing
ChArUco detection +
double TPS denoising
(165 corners × 10 frames)

pixel
correspondences

Zernike Rayfield BA
Per-pixel ray fitting
 $0(u,v)+d(u,v)$, $0 \cdot d=0$
transverse gauge

Physical Model Identification (3D)

Ray-Space BIC
Model selection in ray space
Telecentric CMO wins
($\Delta BIC > 40,000$)

Zernike rayfield
(per-pixel rays)

CMO family +
parameter seed

CMO + SE(3) Model
Joint telecentric fit +
per-channel arm alignment
(26 params, 1.06 px RMS)

OUTPUT

Calibrated rays
+ physical model